

Text Embeddings in NLP

What are Embeddings?

Embeddings convert words or sentences into dense numerical vectors. Similar words end up close together in vector space. They capture semantic meaning.

Types of Embeddings

- Word2Vec: predicts surrounding words (CBOW) or centre word (Skip-gram)
- GloVe: Global Vectors — uses word co-occurrence statistics
- FastText: handles subwords, works on unseen/rare words
- BERT Embeddings: contextual — same word gets different vectors in different sentences

Word2Vec Example

- king - man + woman $\approx$ queen
- paris - france + italy $\approx$ rome
- Vector dimension: typically 100–300

How to Use (Python)

- `from gensim.models import Word2Vec`
- `model = Word2Vec(sentences, vector_size=100, window=5)`
- `vector = model.wv['python']` # get embedding for 'python'

Applications

- Semantic search and document similarity
- Sentiment analysis and text classification
- Machine translation and question answering